Class Activity — 10.07.2026

Data Handling & Visualization: R Code + Output for All 10 Scenarios

R (ggplot2 / dplyr / gridExtra) — hardcoded / built-in datasets

Generated: 13 July 2026

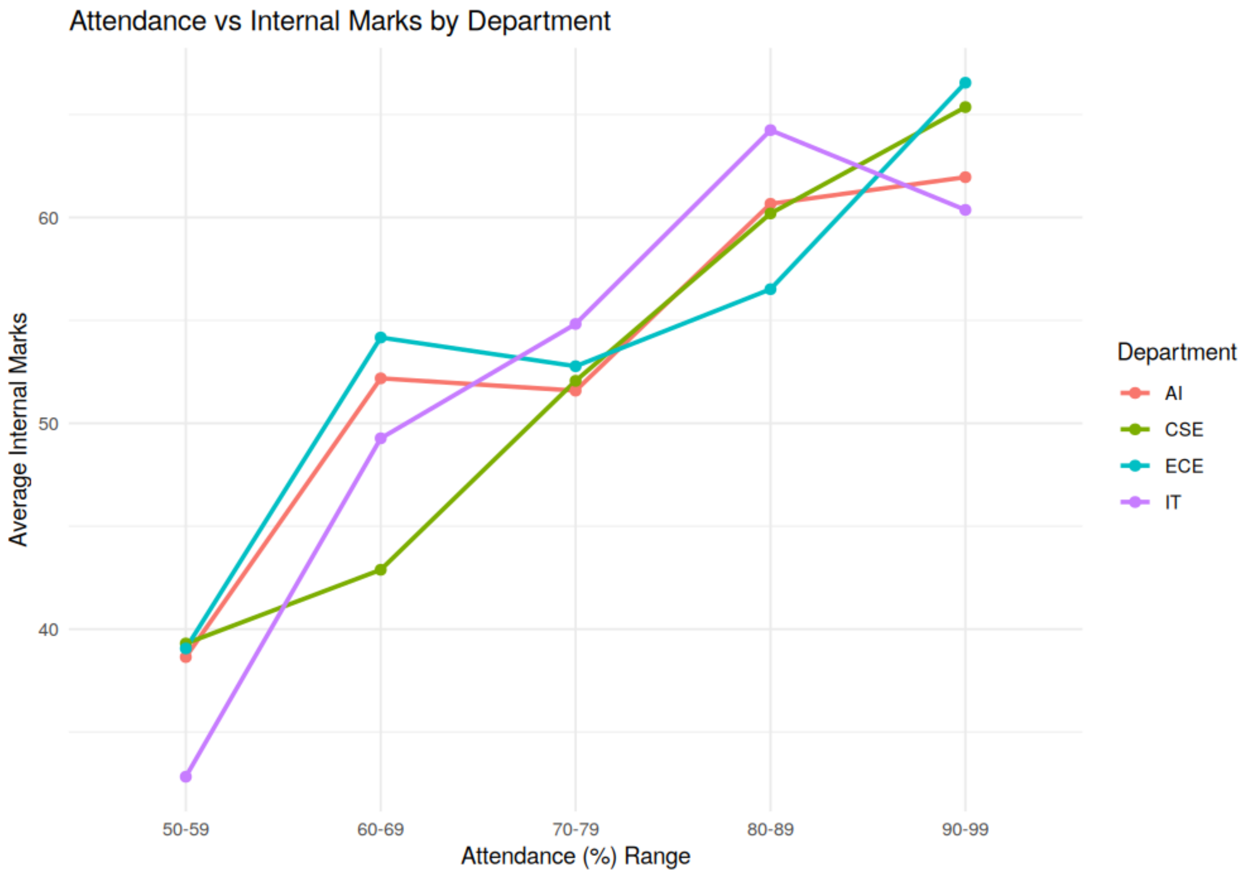
Scenario 1

Student Academic Performance Dashboard

Code (R)

```
library(ggplot2)
library(dplyr)

set.seed(1)
n &lt;- 60
dept &lt;- sample(c('CSE','AI','IT','ECE'), n, replace=TRUE)
att &lt;- sample(50:99, n, replace=TRUE)
marks &lt;- att*0.7 + rnorm(n, 0, 8)
df1 &lt;- data.frame(Attendance=att, Marks=marks, Department=dept)

df1$AttBin &lt;- cut(df1$Attendance, breaks=c(50,60,70,80,90,100),
                   labels=c('50-59','60-69','70-79','80-89','90-99'), right=FALSE)

grouped &lt;- df1 %&gt;% group_by(AttBin, Department) %&gt;% summarise(Marks=mean(Marks), .groups='drop')

ggplot(grouped, aes(x=AttBin, y=Marks, color=Department, group=Department)) +
  geom_line(size=1) + geom_point(size=2) +
  labs(title="Attendance vs Internal Marks by Department",
       x="Attendance (%) Range", y="Average Internal Marks") +
  theme_minimal()
```

Output



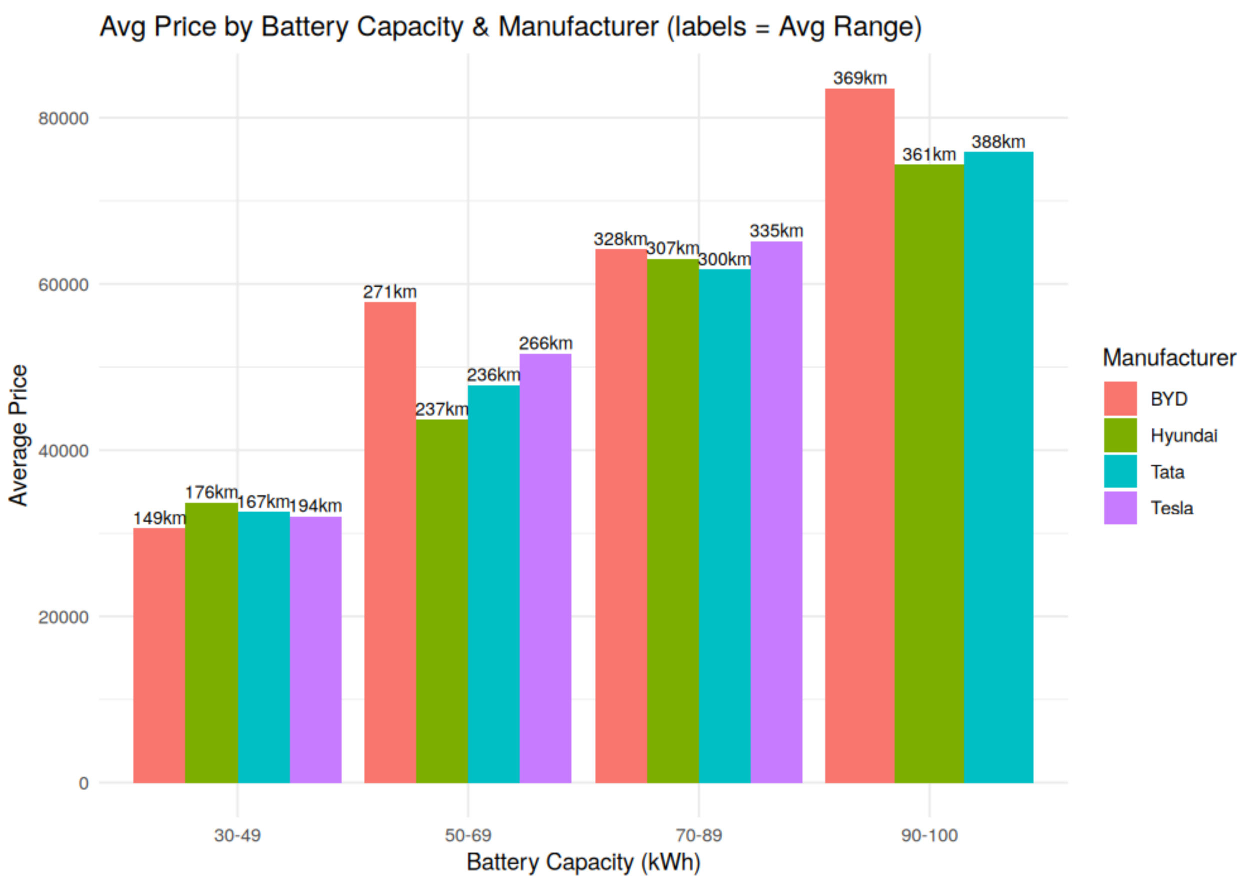
Scenario 2

Electric Vehicle Market Analysis

Code (R)

```
library(ggplot2)
library(dplyr)

set.seed(1)
n &lt;- 40
battery &lt;- runif(n, 30, 100)
mfr &lt;- sample(c('Tesla','Tata','BYD','Hyundai'), n, replace=TRUE)
price &lt;- battery*800 + rnorm(n, 0, 3000)
range_km &lt;- battery*4 + rnorm(n, 0, 20)
df2 &lt;- data.frame(Battery=battery, Price=price, Range=range_km, Manufacturer=mfr)

df2$BatteryBin &lt;- cut(df2$Battery, breaks=c(30,50,70,90,100),
                       labels=c('30-49','50-69','70-89','90-100'), include.lowest=TRUE)

grouped &lt;- df2 %&gt;% group_by(BatteryBin, Manufacturer) %&gt;%
  summarise(AvgPrice=mean(Price), AvgRange=mean(Range), .groups='drop')

ggplot(grouped, aes(x=BatteryBin, y=AvgPrice, fill=Manufacturer)) +
  geom_col(position=position_dodge()) +
  geom_text(aes(label=paste0(round(AvgRange),"km")),
            position=position_dodge(width=0.9), vjust=-0.3, size=3) +
  labs(title="Avg Price by Battery Capacity &amp; Manufacturer (labels = Avg Range)",
       x="Battery Capacity (kWh)", y="Average Price") +
  theme_minimal()
```

Output



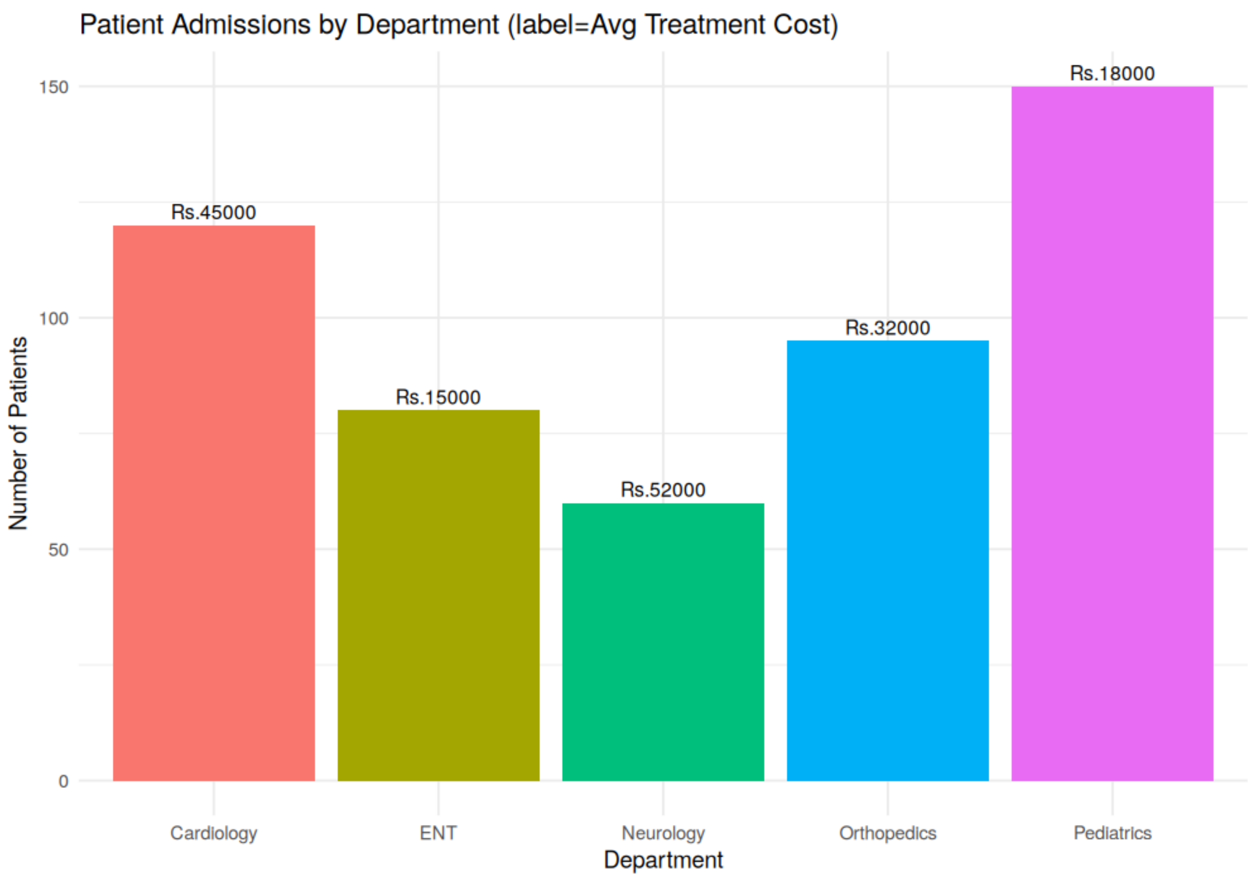
Scenario 3

Hospital Patient Admission Analysis

Code (R)

```
library(ggplot2)

depts &lt;- c('Cardiology','Orthopedics','Pediatrics','Neurology','ENT')
patients &lt;- c(120,95,150,60,80)
cost &lt;- c(45000,32000,18000,52000,15000)
df3 &lt;- data.frame(Department=depts, Patients=patients, Cost=cost)

ggplot(df3, aes(x=Department, y=Patients, fill=Department)) +
  geom_col() +
  geom_text(aes(label=paste0("Rs.", Cost)), vjust=-0.4, size=3.3) +
  labs(title="Patient Admissions by Department (label=Avg Treatment Cost)",
       y="Number of Patients") +
  theme_minimal() +
  theme(legend.position="none")
```

Output



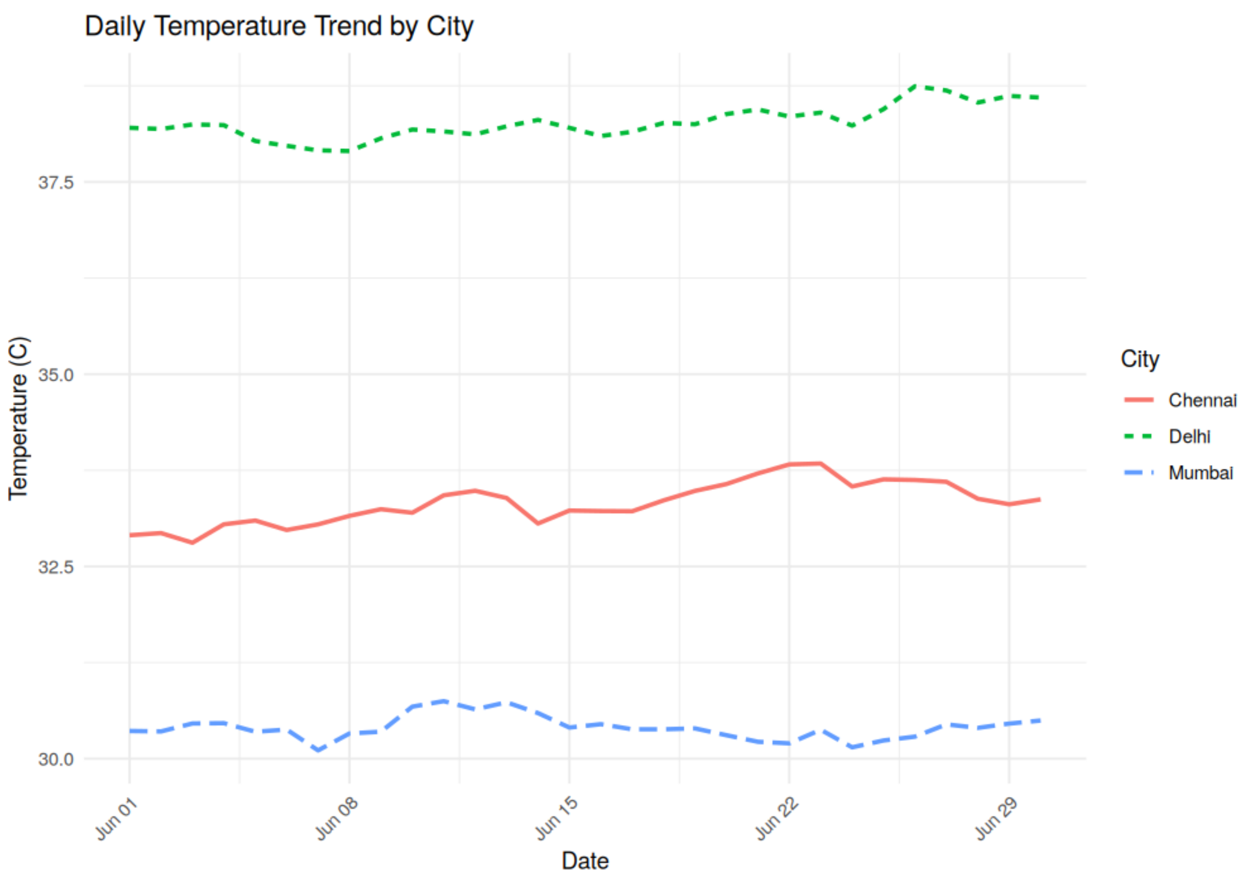
Scenario 4

Weather Monitoring System

Code (R)

```
library(ggplot2)
library(dplyr)

set.seed(1)
days &lt;- seq(as.Date("2026-06-01"), by="day", length.out=30)
cities &lt;- c('Chennai','Delhi','Mumbai')
base_temp &lt;- c(Chennai=33, Delhi=38, Mumbai=30)

df4 &lt;- data.frame()
for (city in cities) {
  temp &lt;- base_temp[[city]] + cumsum(rnorm(30, 0, 1.5))*0.1
  df4 &lt;- rbind(df4, data.frame(Date=days, Temperature=temp, City=city))
}

ggplot(df4, aes(x=Date, y=Temperature, color=City, linetype=City)) +
  geom_line(size=1) +
  labs(title="Daily Temperature Trend by City", y="Temperature (C)") +
  theme_minimal() +
  theme(axis.text.x=element_text(angle=45, hjust=1))
```

Output



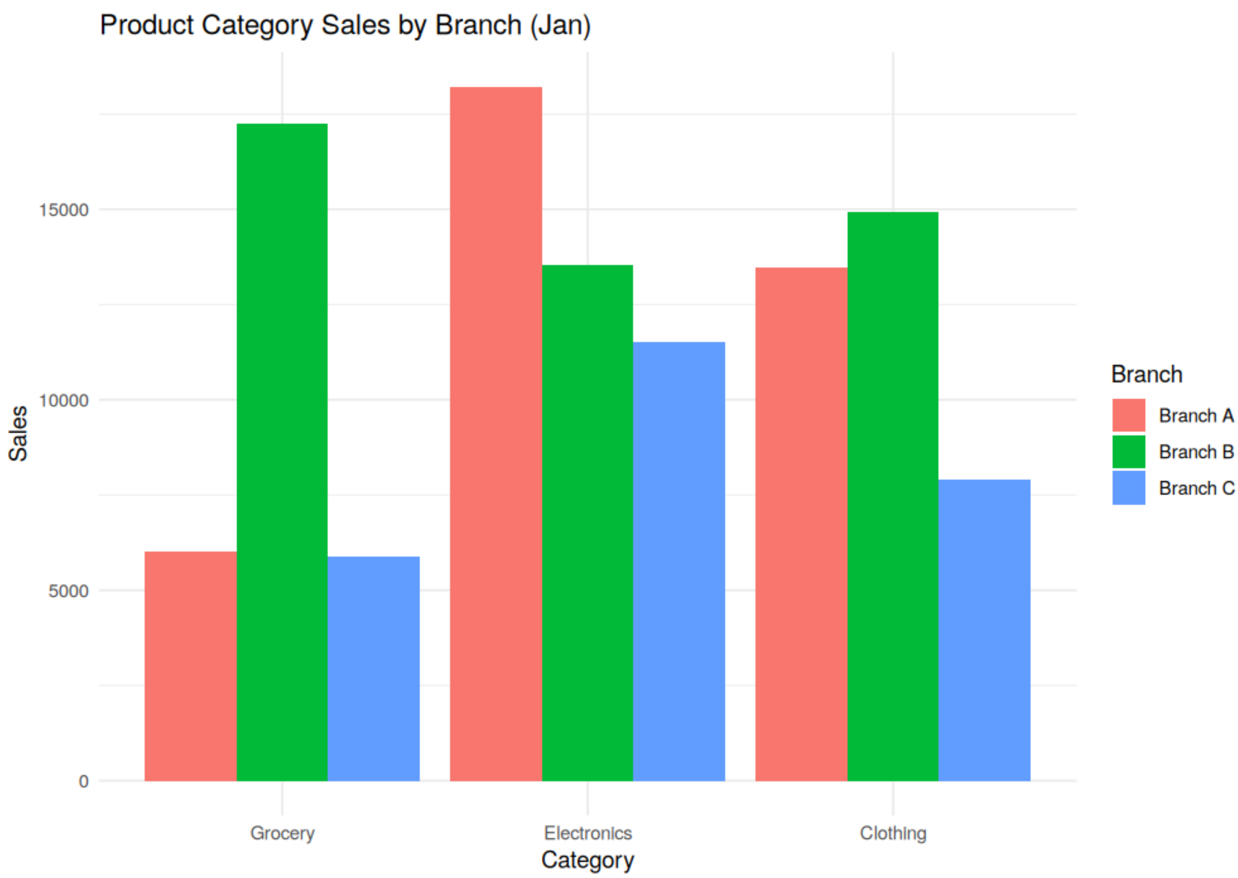
Scenario 5

Supermarket Sales Analysis

Code (R)

```
library(ggplot2)

set.seed(1)
months &lt;- c('Jan','Feb','Mar','Apr')
categories &lt;- c('Grocery','Electronics','Clothing')
branches &lt;- c('Branch A','Branch B','Branch C')

df5 &lt;- expand.grid(Month=months, Category=categories, Branch=branches)
df5$Sales &lt;- sample(5000:20000, nrow(df5), replace=TRUE)

jan_data &lt;- df5[df5$Month=='Jan', ]

ggplot(jan_data, aes(x=Category, y=Sales, fill=Branch)) +
  geom_col(position=position_dodge()) +
  labs(title="Product Category Sales by Branch (Jan)") +
  theme_minimal()
```

Output



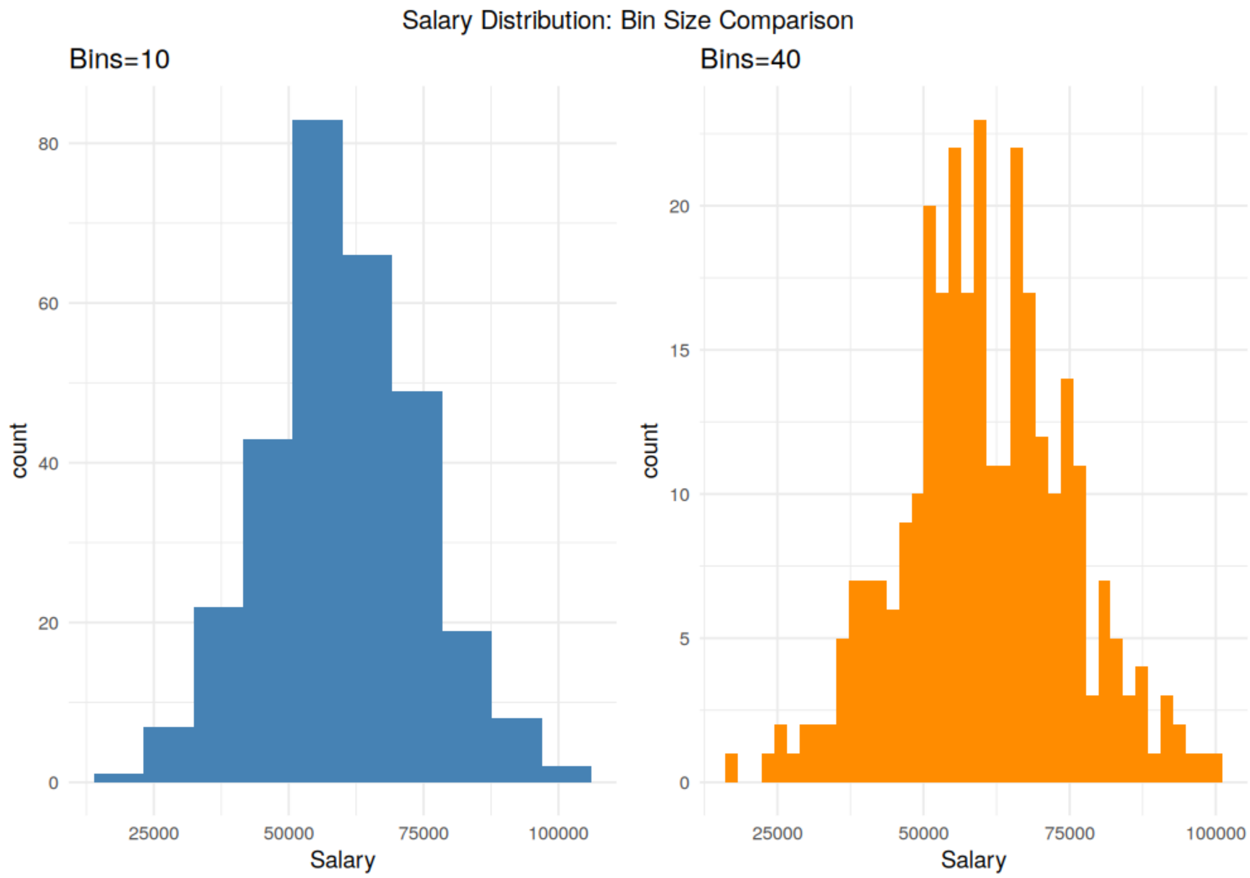
Scenario 6

Employee Salary Distribution

Code (R)

```
library(ggplot2)
library(gridExtra)

set.seed(1)
salary &lt;- rnorm(300, 60000, 15000)
df6 &lt;- data.frame(Salary=salary)

p1 &lt;- ggplot(df6, aes(x=Salary)) +
  geom_histogram(bins=10, fill="steelblue") +
  labs(title="Bins=10") + theme_minimal()

p2 &lt;- ggplot(df6, aes(x=Salary)) +
  geom_histogram(bins=40, fill="darkorange") +
  labs(title="Bins=40") + theme_minimal()

grid.arrange(p1, p2, ncol=2, top="Salary Distribution: Bin Size Comparison")
```

Output



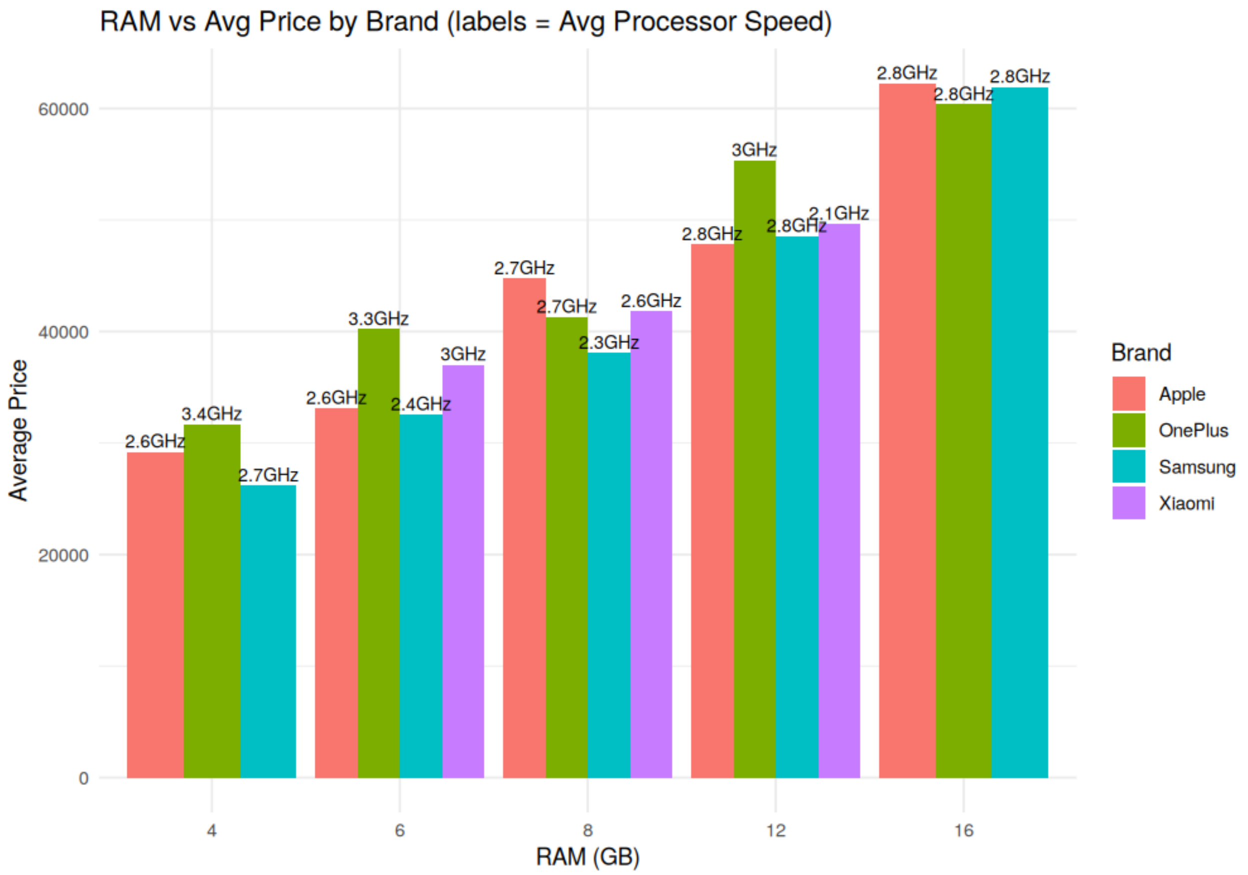
Scenario 7

Smartphone Market Comparison

Code (R)

```
library(ggplot2)
library(dplyr)

set.seed(1)
n &lt;- 50
brand &lt;- sample(c('Samsung','Apple','OnePlus','Xiaomi'), n, replace=TRUE)
ram &lt;- sample(c(4,6,8,12,16), n, replace=TRUE)
storage &lt;- sample(c(64,128,256,512), n, replace=TRUE)
speed &lt;- runif(n, 2.0, 3.5)
price &lt;- ram*3000 + storage*10 + speed*5000 + rnorm(n, 0, 3000)
df7 &lt;- data.frame(Brand=brand, RAM=ram, Storage=storage, Speed=speed, Price=price)

grouped &lt;- df7 %&gt;% group_by(RAM, Brand) %&gt;%
  summarise(AvgPrice=mean(Price), AvgSpeed=mean(Speed), .groups='drop')

ggplot(grouped, aes(x=factor(RAM), y=AvgPrice, fill=Brand)) +
  geom_col(position=position_dodge()) +
  geom_text(aes(label=paste0(round(AvgSpeed,1),"GHz")),
            position=position_dodge(width=0.9), vjust=-0.3, size=3) +
  labs(title="RAM vs Avg Price by Brand (labels = Avg Processor Speed)",
       x="RAM (GB)", y="Average Price") +
  theme_minimal()
```

Output



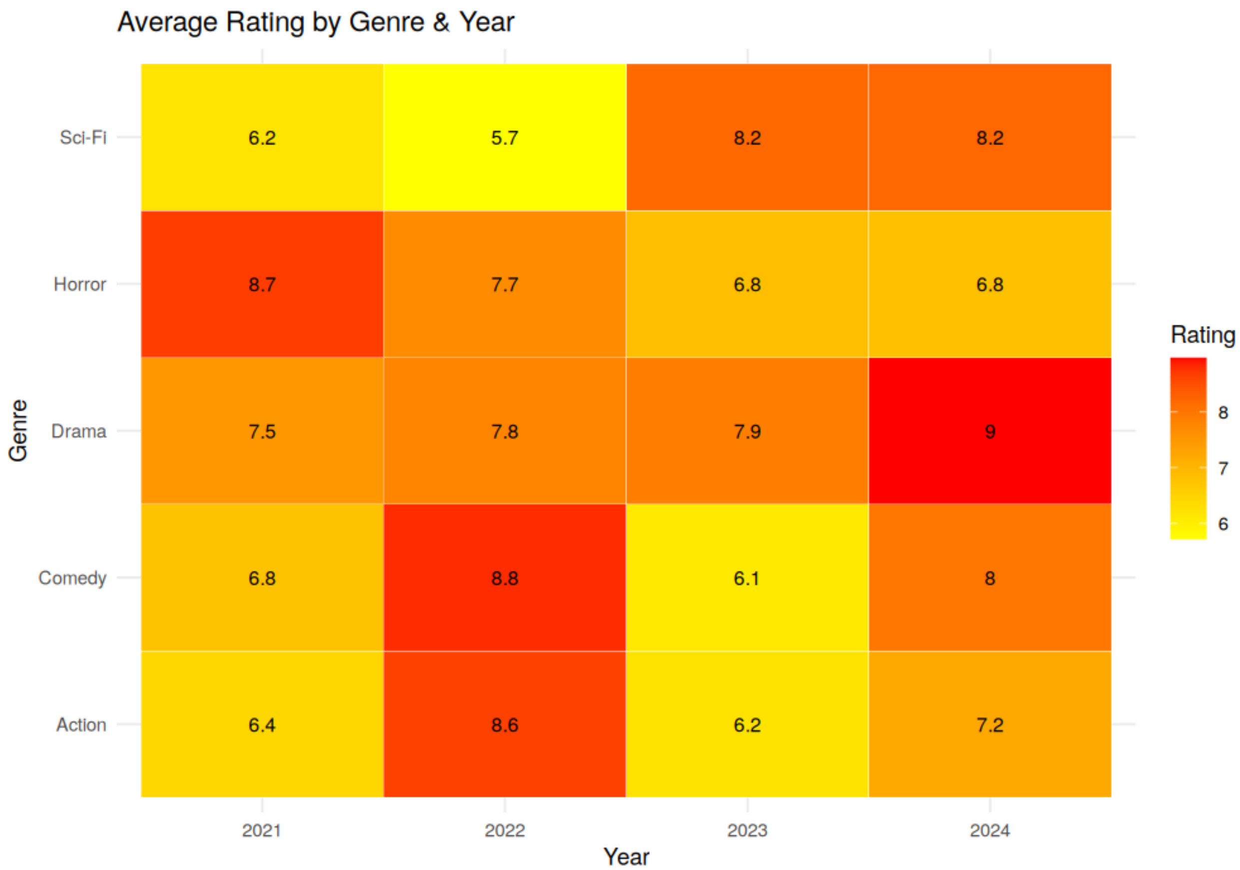
Scenario 8

Online Movie Rating Analysis

Code (R)

```
library(ggplot2)

set.seed(1)
genres &lt;- c('Action','Comedy','Drama','Horror','Sci-Fi')
years &lt;- c(2021,2022,2023,2024)

df8 &lt;- expand.grid(Genre=genres, Year=years)
df8$Rating &lt;- runif(nrow(df8), 5.5, 9.0)

ggplot(df8, aes(x=factor(Year), y=Genre, fill=Rating)) +
  geom_tile(color="white") +
  geom_text(aes(label=round(Rating,1)), size=3.3) +
  scale_fill_gradient(low="yellow", high="red") +
  labs(title="Average Rating by Genre &amp; Year", x="Year", y="Genre") +
  theme_minimal()
```

Output



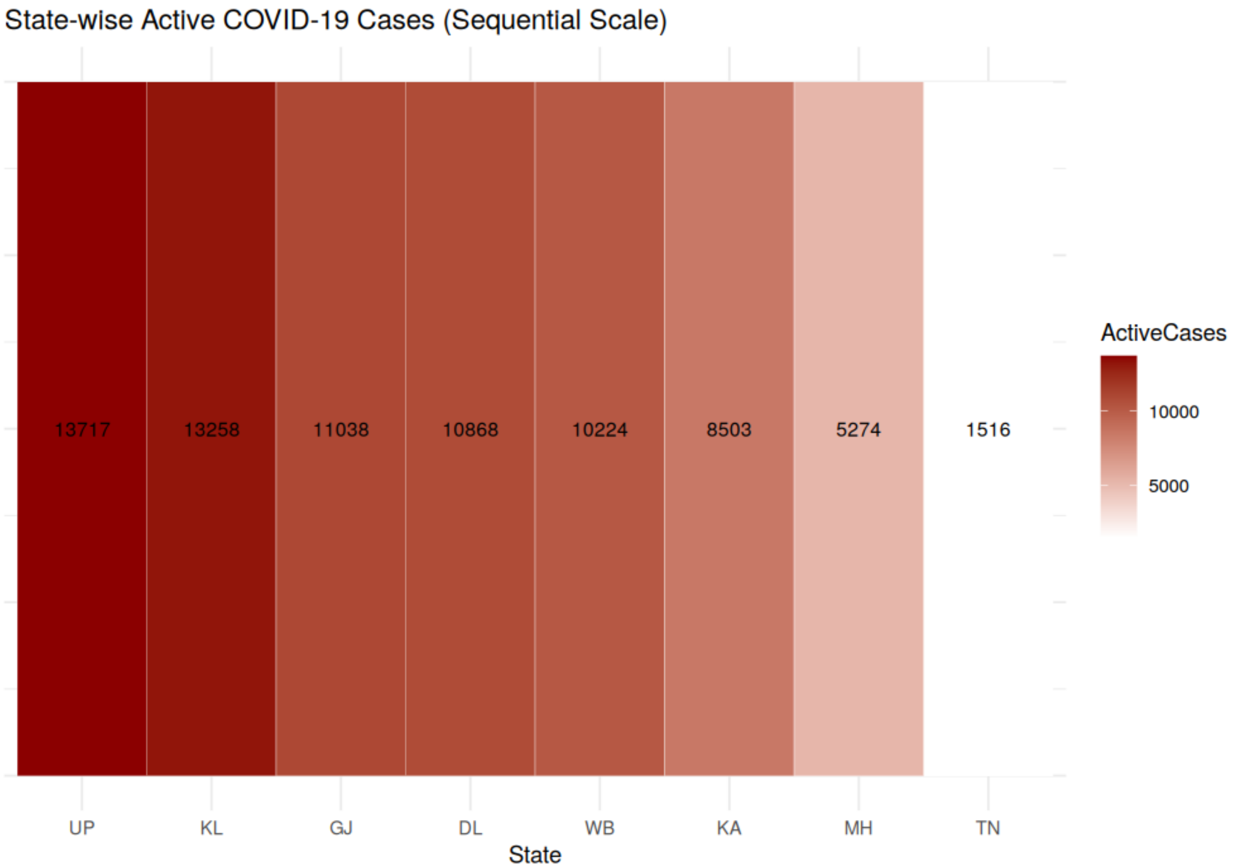
Scenario 9

COVID-19 State-wise Analysis

Code (R)

```
library(ggplot2)
library(dplyr)

set.seed(1)
states &lt;- c('TN','KA','MH','DL','UP','WB','KL','GJ')
active_cases &lt;- sample(500:15000, length(states))
df9 &lt;- data.frame(State=states, ActiveCases=active_cases) %&gt;%
  arrange(desc(ActiveCases))
df9$State &lt;- factor(df9$State, levels=df9$State)

ggplot(df9, aes(x=State, y=1, fill=ActiveCases)) +
  geom_tile(color="white") +
  geom_text(aes(label=ActiveCases), size=3.3) +
  scale_fill_gradient(low="white", high="darkred") +
  labs(title="State-wise Active COVID-19 Cases (Sequential Scale)") +
  theme_minimal() +
  theme(axis.title.y=element_blank(), axis.text.y=element_blank())
```

Output



Scenario 10

Iris Flower Classification

Code (R)

```
library(ggplot2)
library(gridExtra)

data(iris)

p1 &lt;- ggplot(iris, aes(x=Species, y=Sepal.Length, fill=Species)) +
  geom_boxplot() +
  labs(title="Sepal Length by Species") +
  theme_minimal() + theme(legend.position="none")

p2 &lt;- ggplot(iris, aes(x=Species, y=Petal.Length, fill=Species)) +
  geom_boxplot() +
  labs(title="Petal Length by Species") +
  theme_minimal() + theme(legend.position="none")

grid.arrange(p1, p2, ncol=2, top="Species Separation via Sepal/Petal Length (Box Plots)")
```

Output